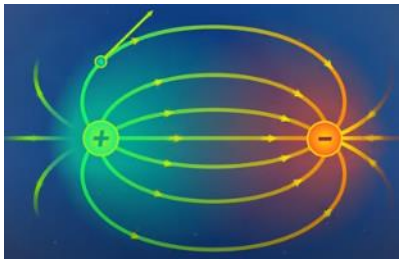
EE4047 – High Voltage Techniques I

New Topic:
Electrostatic Fields and Planar Electrode Systems



Week 5.2 – Thursday Lesson

Asst. Prof. Dr. M. Alparslan Zehir

30.03.2023

OUTLINE

1 Electrostatic Field

2 Electrostatic Field in Planar Electrode Systems

1 ELECTROSTATIC FIELD

The field caused by nonmoving electric charges is called as **electrostatic field**.

The field caused by the electric charges that move with a constant speed is called as **magnetostatic field**.

The fied caused by accelerating electric charges is **electromagnetic field**, in which electric and magnetic fields are changing by time.

1 ELECTROSTATIC FIELD

In high voltage technique, in order to investigate flashover and breakdown events and the behavior of equipment and material under high voltage, electrostatic field calculations are needed.

Scalar and Vectoral Quantities

The quantities independent from a coordinate system are called as **scalar quantities**.

The quantities that are determined with a magnitude and direction are called as **vectoral quantities**.

1 ELECTROSTATIC FIELD

Generally, physical quantities change as a function of **space** or **time**.

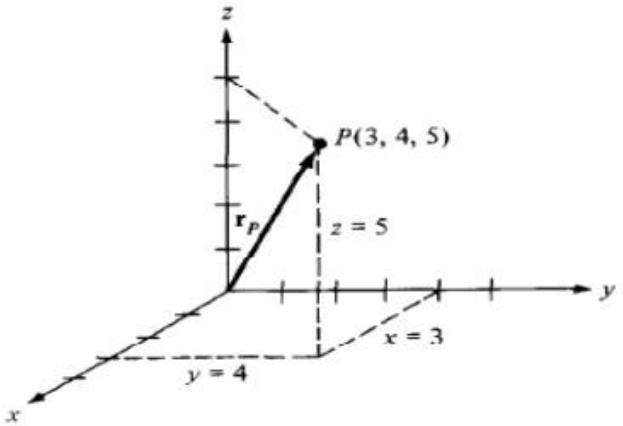
In order to express these changes, the **points in space** should be defined properly.

This requires using a **coordinate system**.

A point or a vector can be represented in an **orthogonal system** in which all the coordinates are **perpendicular to each other**. Orthogonal coordinate systems can be **cartesian** (**rectangular**), **circular cylindrical**, **spherical** and other.

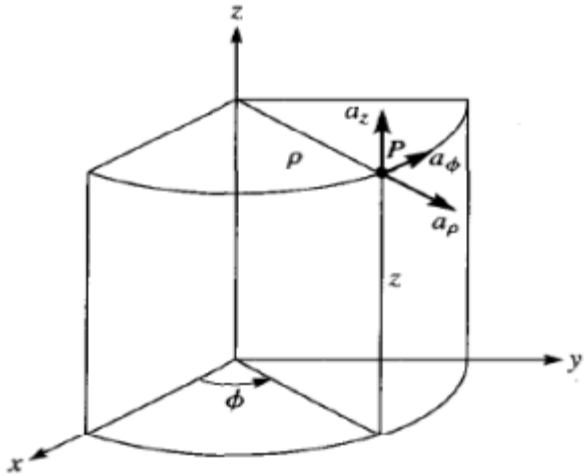
ELECTROSTATIC FIELD

$$-\infty < x < \infty$$
$$-\infty < y < \infty$$
$$-\infty < z < \infty$$



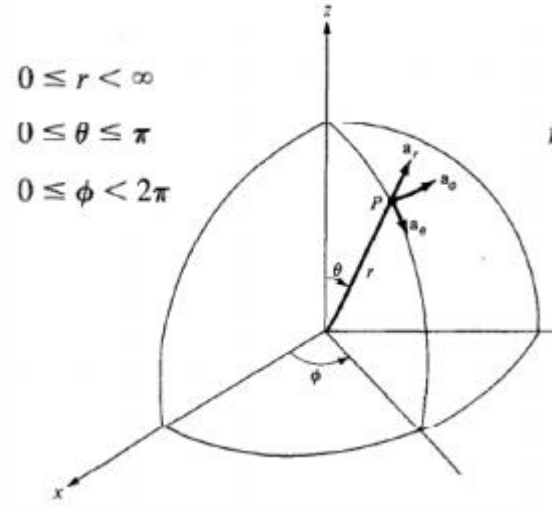
Point P and position vector

$$0 \le \rho \le \infty$$
$$0 \le \phi \le 2\pi$$
$$-\infty \le z \le \infty$$



Point P and unit vectors in the cylindrical coordinate system.

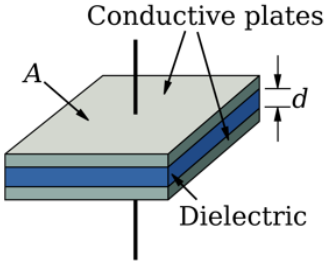
$$0 \le r < \infty$$
$$0 \le \theta \le \pi$$
$$0 \le \phi < 2\pi$$



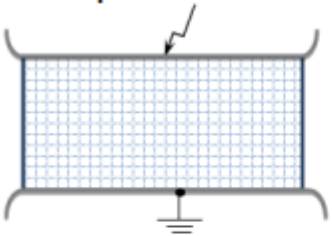
Point P and unit vectors in spherical coordinates.

ELECTROSTATIC FIELD

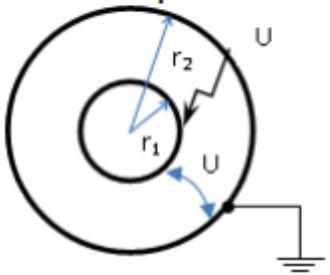
Basic electrode systems:



Parallel plate electrodes

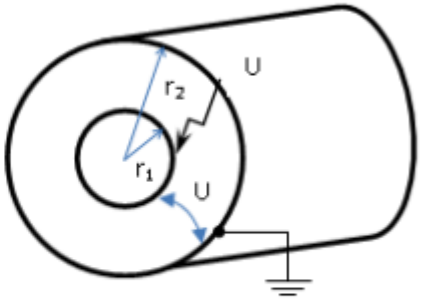


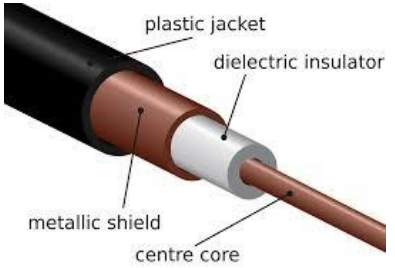
- Concentric spheres



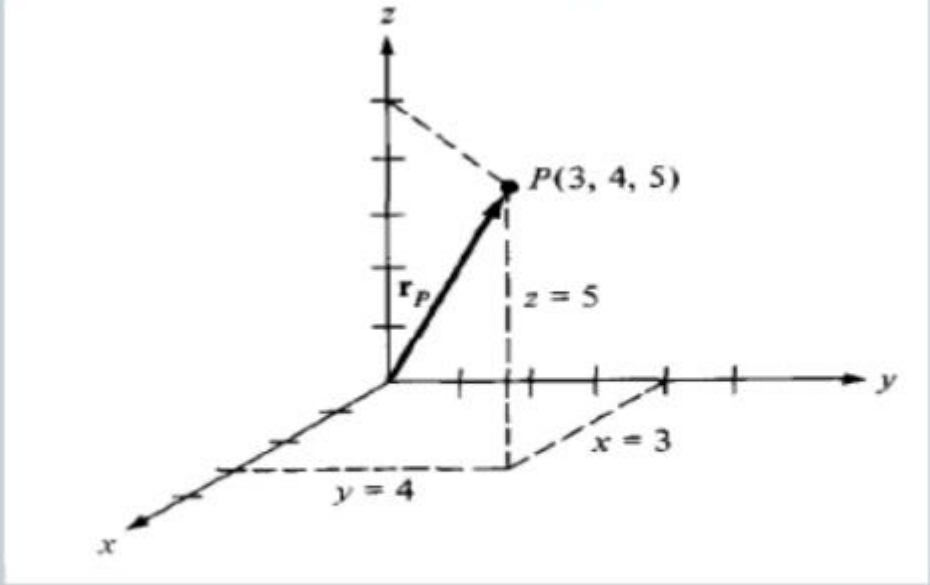


- Co-axial cylinders





Cartesian Coordinate System



$$-\infty < x < \infty$$
$$-\infty < y < \infty$$
$$-\infty < z < \infty$$

Figure 1.4 Illustration of position vector $\mathbf{r}_P = 3\mathbf{a}_x + 4\mathbf{a}_y + 5\mathbf{a}_z$.

Cartesian Coordinate System

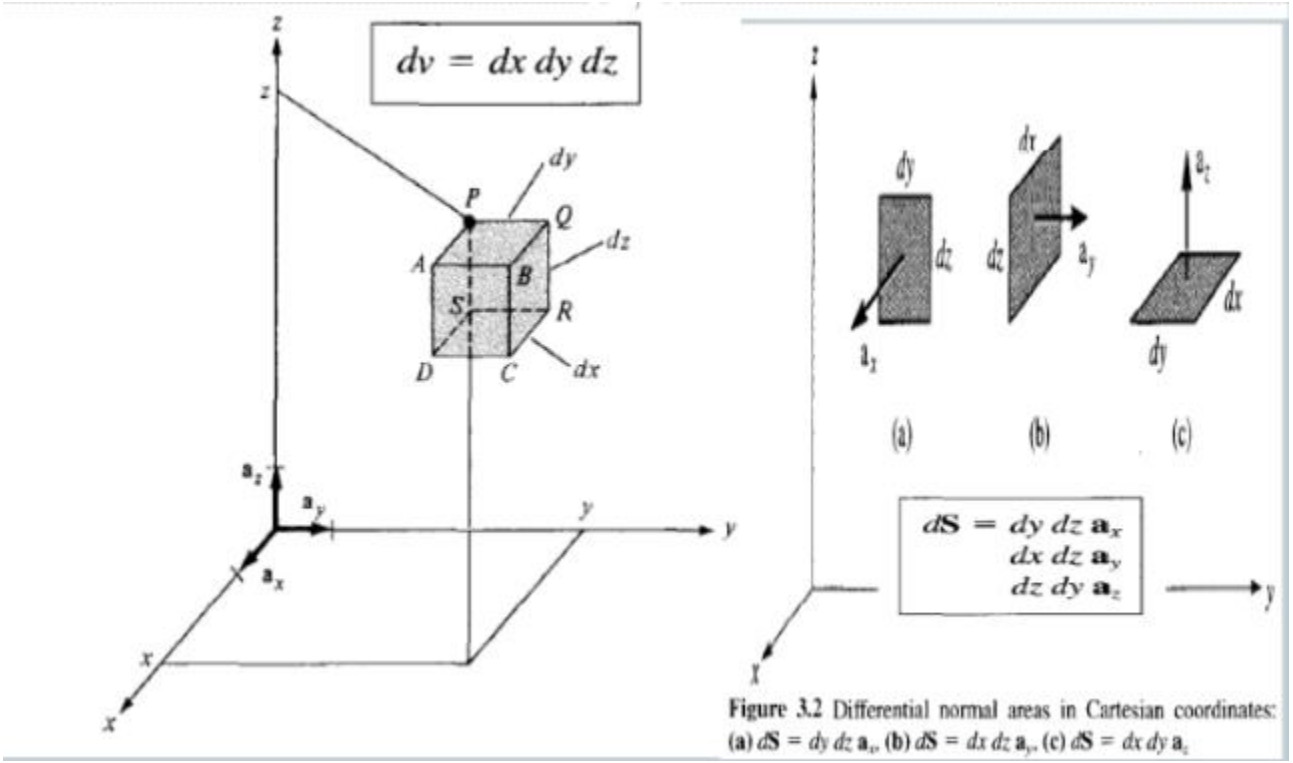


Figure 3.2 Differential normal areas in Cartesian coordinates: (a) $dS = dy\,dz\,a_x$, (b) $dS = dx\,dz\,a_y$, (c) $dS = dx\,dy\,a_z$

Scalar Multiplication

The **scalar multiplication of two vectors** such as $\vec{A}$ and $\vec{B}$, is the multiplication of their magnitudes with the cosine of the angle between them.

$$\vec{A}.\vec{B} = |A||B|.cos\theta_{AB}$$

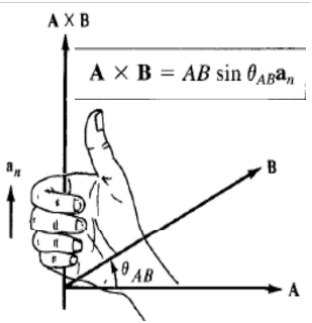
Vectoral Multiplication

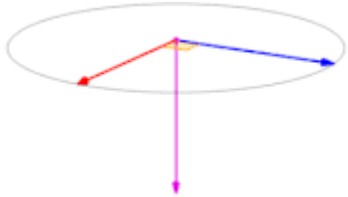
The vectoral multiplication of two vectors such as $\vec{A}$ and $\vec{B}$, is shown as $\vec{A} \times \vec{B}$. The result is another vector, magnitude of which is equal to the multiplication of the maginitudes of $\vec{A}$ and $\vec{B}$ and the sine of the angle between them, and direction of which is perpendicular to the plane of $\vec{A}$ and $\vec{B}$ vectors.

$$\vec{A} = A_x.i + A_y.j + A_z.k$$

$$\vec{B} = B_x.i + B_y.j + B_z.k$$

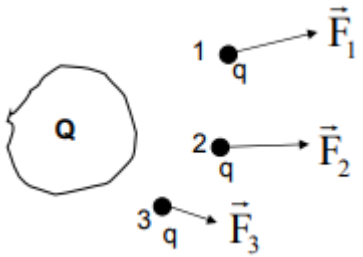
$$\vec{A} \times \vec{B} = \begin{vmatrix} i & j & k \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$





ELECTROSTATIC FIELD

The impact of an object with charge Q on a particle with charge q:



$$\vec{F} = k.q.\vec{E}$$

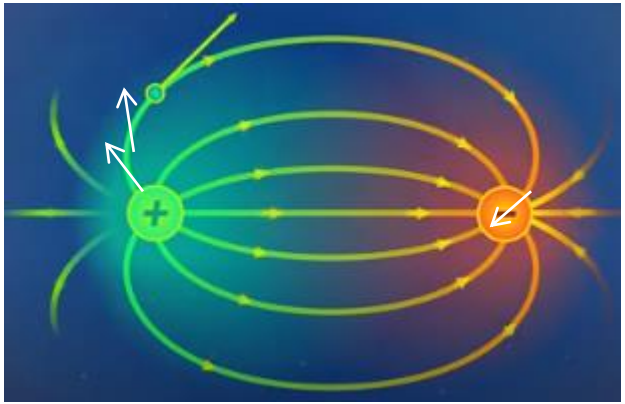
If the unit of q is C, electric field unit is V/m and the force is N, k=1

$$\vec{F} = q.\vec{E}$$

$$\vec{E} = \frac{\vec{F}}{q}$$

The field lines are tangent to the direction of the field at any point.

The greater the line density the greater the magnitude of the field.

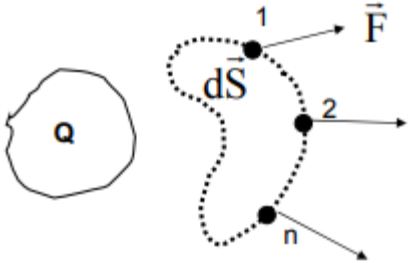


The lines always start from positiely charged object and end on negatively charged objects.

The lines must never cross.

ELECTROSTATIC FIELD

If a particle with q electric charge is moved through a closed curve (square, circle, ellipsis and other) the net work is zero.



This is regardles of the shape of the path the particle followed.

It has to be a closed path and the particle has to reach the point again. This is regardless of the electric field

$$\oint \vec{F} d\vec{S} = 0$$

$$\oint q\vec{E} d\vec{S} = 0$$

$$\oint q\vec{E} d\vec{S} = \int (rot\vec{E})\, d\vec{S} = 0$$

$$rot\vec{E} = 0$$

$\nabla = \frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k$ (gradient, taking derivatives in each dimension)

$$\vec{E} = -\nabla . V = -grad . V$$

Laplace and Poisson Equations

Electric charge in a closed plane is equal to the electric flux leaving the plane.

$$\varepsilon_0 \oint \vec{E} d\vec{S} = Q$$

In closed plane involving insulating material, the same applies as

$$\varepsilon_0 \varepsilon_R \oint \vec{E} d\vec{S} = Q$$

$$\oint \varepsilon_0 \varepsilon_R \vec{E} d\vec{S} = Q$$

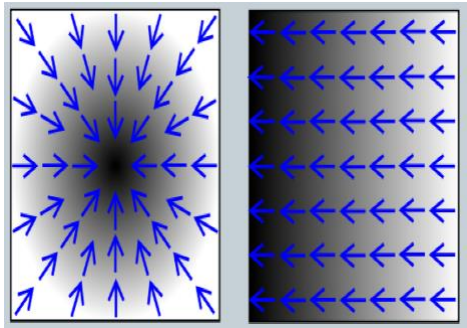
$\varepsilon = \varepsilon_0 \varepsilon_R$ $\varepsilon_0 = 8{,}86 \times 10^{-12} F/m$ (permittivity of vacuum)

$\vec{D} = \varepsilon \vec{E}$

$\vec{D}$ is electric flux density. ε is dielectric constant.

$\oint \vec{D} d\vec{S} = Q$ $\Psi = \int \vec{D} d\vec{S} = Q$

ELECTROSTATIC FIELD



$$\nabla \vec{E} = \frac{\rho}{\varepsilon}$$

$$\nabla \vec{D} = \rho$$

Substituting $\vec{E} = -\nabla . V$

$$\nabla(\nabla V) = -\frac{\rho}{\varepsilon} \qquad \mathrm{div}(gradV) = -\frac{\rho}{\varepsilon}$$

$$\nabla^2 V = -\frac{\rho}{\varepsilon} \quad \text{(Poisson)}$$

$$\nabla^2 V = 0 \quad \text{(Laplace)}$$

The gradient of a scalar area gives a vectoral area directed towards the change.

Divergance is the flux per unit mass, when mass goes to zero.

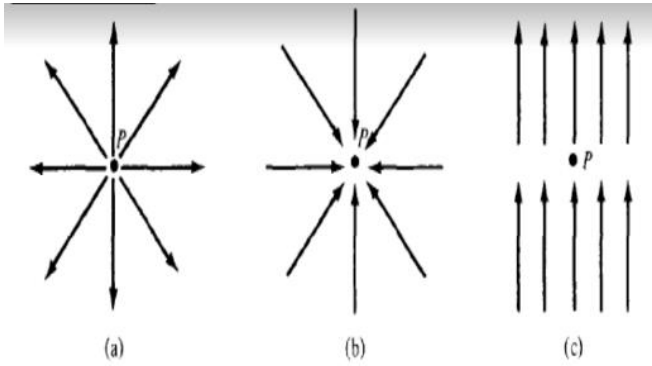
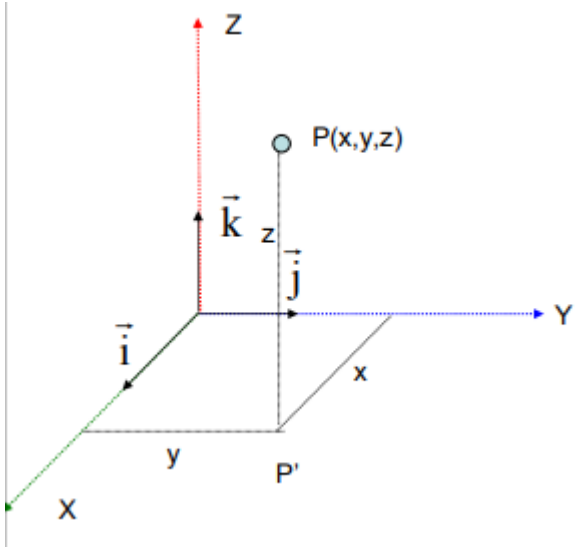


Figure 3.14 Illustration of the divergence of a vector field at *P*; **(a)** positive divergence, **(b)** negative divergence, **(c)** zero divergence.

Cartesian Coordinate System

Laplacian $\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$

Gradient $\nabla V = \frac{\partial V}{\partial x} i + \frac{\partial V}{\partial y} j + \frac{\partial V}{\partial z} k$



In case the plane changes through the x axis

$\frac{d^2 V}{dx^2} = 0$ $E = -\frac{dV}{dx}$

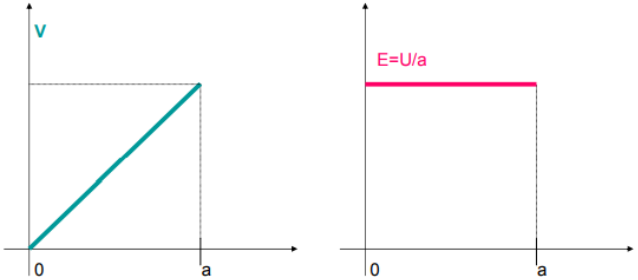
ELECTROSTATIC FIELD IN PLANAR ELECTRODE SYSTEMS

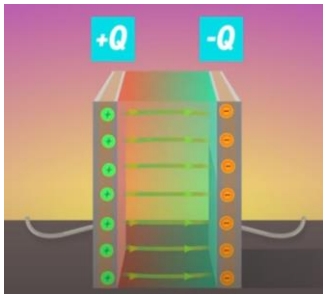
Planar Electrode Systems (without corners, infinite length)

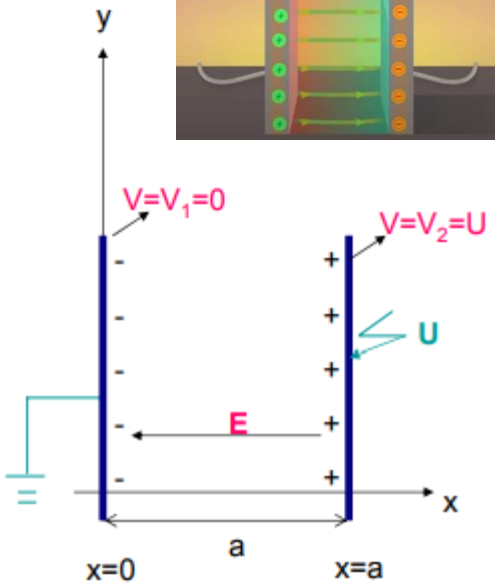
$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

$$\frac{\partial^2 V}{\partial x^2} = 0 \quad ; \quad V = \frac{U}{a}x \quad ; \mathrm{E} = -\frac{U}{a}$$

Capacity: $\mathrm{C} = \frac{Q}{U} = \frac{\varepsilon S}{a}$ F







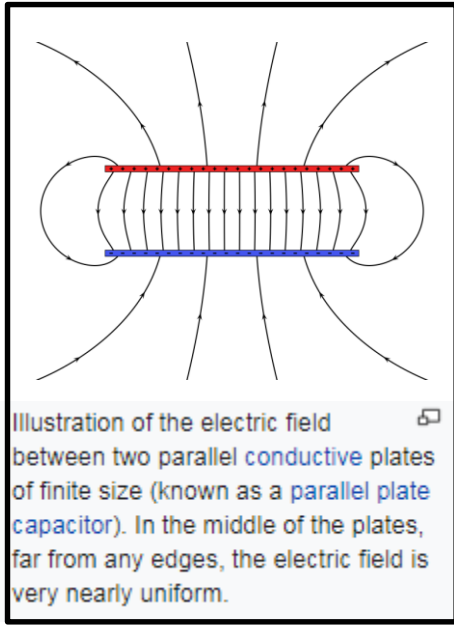


Illustration of the electric field between two parallel conductive plates of finite size (known as a parallel plate capacitor). In the middle of the plates, far from any edges, the electric field is very nearly uniform.

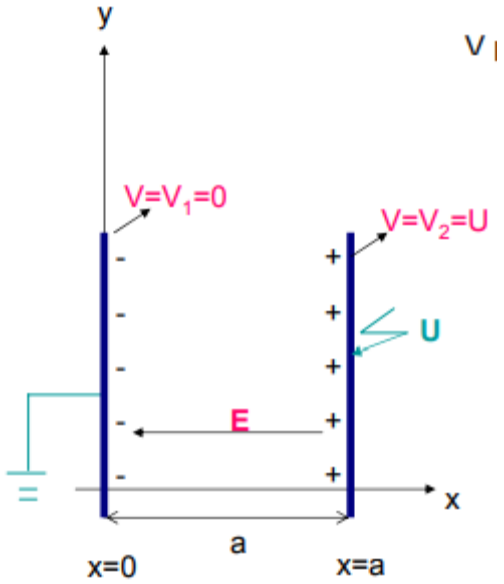
Planar Electrode Systems (without corners, infinite length)

If electrical stress is equal to or higher than dielectric strength discharge starts.

$E \geq E_d$ (kV/cm)

In non-uniform fields, first partial discharge events occur and if voltage is further increased, full breakdown is encountered.

In uniform fields, every point has the same stress and discharge in one electron ends up as full breakdown.



Therefore, in uniform fields, the voltage at discharge starts and the voltage causes full breakdown is the same. Breakdown voltage can be found as $U_d = E_d . a$

Example-1:

In a planar electrode system, electrode gap is a=2 mm, electrode surface area $S = 4\,cm^2$, $\varepsilon_r = 1\,(air)$ and the applied voltage is U=5 kV. Calculate the capacity, electric charge, electric field strength, electric flux density and the maximum voltage that can be applied if the dielectric strenght of the insulation material is 30 kV/cm. ($\varepsilon_0 = 8{,}854.10^{-12}\,\frac{F}{m}$)

Solution: Capacity of the electrode system:

$$C = \frac{\varepsilon S}{a} = \frac{\varepsilon_0 \varepsilon_r S}{a} = \frac{8{,}854.10^{-12}.1.4.10^{-4}}{2.10^{-3}} = \frac{F.m^2}{m.m} = 1{,}7708.10^{-12}\,F = 1.7708\,pF$$

Electric charge:

$$\mathrm{Q} = C.U = 1{,}7708.10^{-12}.5.10^{3} = 8.854.10^{-9}\,C = 8.854\,nC$$

Electric field strength:

$$E = \frac{U}{a} = \frac{5\,kV}{2.10^{-1}} = 25\,kV/cm = 2500\,\frac{kV}{m} = 25.10^{5}\,\frac{V}{m}$$

Formulas:

$$D = \varepsilon E$$

$$E = \frac{U}{a}$$

$$\mathrm{Q} = C.U$$

$$C = \frac{\varepsilon S}{a}$$

Example-1:

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Continuation to Solution: Electric flux density:

$$D = \varepsilon E = \varepsilon_0 \varepsilon_R E = 8{,}854.10^{-12}.1.25.10^5 = 2.2135.10^{-5}\,C/m^2$$

The breakdown voltage:

$$U_d = E_d.a = 30\,\frac{kV}{cm}.2.10^{-1}cm = 6\,kV$$

The maximum voltage that can be applied:

$$\mathrm{U} < U_d = 6\,kV$$

END OF THE LESSON

Any questions?

Check out our playlists on Youtube:

High Voltage Technique: https://www.youtube.com/playlist?list=PLLZI7WaEmZi04KMWVURGlU34W3LoNZyWX

Insulators: https://www.youtube.com/playlist?list=PLLZI7WaEmZi1ReLGq0qfP1VjRqxDJi5kA

Lightning Impulse Voltage: https://www.youtube.com/playlist?list=PLLZI7WaEmZi2isWOB-zBbb6LNplH28byQ